

number			
7(a)	• correct calculation (1) • 7.56 (1)	$340 \div 45 = 7.555555555556$ e.g. 8 / 7.6 / 7.556 / 7.5556 etc ACCEPT ecf 8.29 Bald answer of 7.56 = 2 marks Bald answer of 8 / 7.6 / 7.556 / 7.5556 etc / 8.29 = 1 mark	(2)

Question number	Answer
*7(b)	Table 1: • RYR decreases total cholesterol • RYR decreases LDL • studies used different doses and decreases were not consistent • which suggests other factors may be involved • no indication of sample size / information about sex at birth, age, diet / where levels were measured, when levels were measured • therefore <u>valid</u> conclusions cannot be drawn • credit effect of limitation eg if person ate more cholesterol then decrease might be higher, levels measured closer to meal might be higher • discrepancy on effects of HDL / not much data on HDL • LDL : HDL ratio cannot be calculated • and this is more significant than LDL / total cholesterol levels • no indication of repeats • so conclusions will not be <u>valid</u> • no control group • so cannot be certain that the RYR caused the changes Graph 1: • RYR lowers LDL at both doses • levels of LDL increase once RYR not taken anymore • small sample size • no indication of e.g. diet, age, sex at birth, when and where levels were measured • therefore <u>valid</u> conclusions cannot be drawn • credit effect of limitation eg if person ate more cholesterol then decrease might be higher, levels measured closer to meal might be higher • {large / overlapping} {range bars / error bars} • so conclusions will not be <u>valid</u> / RYR may have no significant effect • short term study • so don't know if effects are long term Graph 2: • taking RYR appears to decrease LDL • {large / overlapping} {range bars / error bars} • so conclusions will not be <u>valid</u> / RYR may have no significant effect • no indication what the , what the dose was, what the time scale was • credit specific effect of limitation eg if person ate more cholesterol then decrease might be higher, levels measured closer to meal might be higher, no timescale no indication of how long drop in LDL took NB A specific limitation / implication can only be awarded once

			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	Demonstrates isolated elements of biological knowledge and understanding to the given context with generalised comments made. Vague statements related to consequences are made with limited linkage to a range of scientific ideas, processes, techniques and procedures. The discussion will contain basic information with some attempt made to link knowledge and understanding to the given context.	Descriptions of data 1 mark = one description 2 marks = at least two descriptions of data from two different sources of data
Level 2	3-4	Demonstrates adequate knowledge and understanding by selecting and applying some relevant biological facts / concepts. Consequences are discussed which are occasionally supported through linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows some linkages and lines of scientific reasoning with some structure.	Limitations of data 3 marks = one limitation described 4 marks = at least two limitations described from two different sources of data
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting and applying relevant biological facts / concepts. Consequences are discussed which supported throughout by sustained linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows a well-developed and sustained line of scientific reasoning which is clear and logically structured.	Implications of limitation 5 marks = one implication of one limitation given 6 marks = at least one implication described for two limitations described from two different sources of data

number			
7(c)(i)	• credit two from: headache pins and needles hair loss acne allergic reaction / rash / hives dizziness memory loss / confusion feeling sick / nausea / vomiting feeling unusually tired / fatigue / physically weak stomach pain digestive system problems, such as constipation, diarrhoea, indigestion or farting loss of appetite (type II) diabetes impotence muscle {pain / tiredness / weakness / damage} joint {pain / swelling} tendon problems sleep problems low blood platelet count liver damage / hepatitis / yellowing of {eyes / skin} / dark-coloured urine kidney failure		(1)

number			
7(c)(ii)	A description that makes reference to four of the following points: • large group of people (1) • people who are {healthy / have no known illnesses} (1) • {control / same} {age / sex at birth / ethnicity / lifestyle / etc} (1) • one group given placebo and one given monacolin K (1) • monitored for side effects (1) • statistical analysis of data (on side effects) / T test (1)	ACCEPT minimum size of 100 ACCEPT not taking other medications DO NOT ACCEPT patients DO NOT ACCEPT contradictions ACCEPT given a control (substance) IGNORE use of different doses ACCEPT over a period of time minimum of a month ACCEPT analysis of Mann-Whitney U / Z test / standard deviation	(4)

number			
7(c)(iii)	An answer that makes reference to two of the following points: • difficult to eliminate the effects of a named factor that cannot be controlled (1) • psychological effects / placebo effect / assumption that they are taking monacolin (1) • cannot quantify the extent of the side effect (1) • different people may react differently to the drug (1) • ethics of giving someone an untested drug (1) • some side effects may {be rare / occur after a long period of time / not be detected} (1)	e.g. diet, activity levels ACCEPT assessment of side effect is subjective ACCEPT people are genetically different side effects may not affect the group of people in the study ACCEPT moral / not right for ethical side effects could be fatal	(2)